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Effects of soret diffusion on the laminar flame speed of *n*-butane-air mixtures

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Abstract

The effect of Soret diffusion on flame speed of the freely-propagating planar premixed flames was computationally studied for *n*-butane/air mixtures using a detailed reaction mechanism and transport. The results show that Soret effect almost has no influence on the *n*-butane/air flame speed, and the sensitivity analysis of flame speed shows that only very light species H radical is involved in the ranked key reactions. That is to say, the Soret diffusion of H radical has no influence on the flame speed. However, the Soret effect of H plays an important role on hydrogen/air flame speed. We found that the Soret diffusion flux of the kth species just exists in the overlap region which consists of the rear part of "driving force" gradT/T profile and the front part of mole fraction profile of the kth species in the freely-propagating planar premixed flame. Compared with hydrogen/air flame, the investigation shows that for the *n*-butane/air flame, due to the inherent feature of total heat release rate, the high temperature largely reduces the "driving force" gradT/T especially on the rear part of its profile and compared with the peak of temperature gradient the location of "driving force" peak is closer to the unburned side, on the other hand the broader consumption layer of H weakens the front part of mole fraction of H and makes the peak of H mole fraction far away from the unburned side. Thus, the overlap region is narrow and low values of "driving force" and mole fraction of H embed in it.
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Keywords: Soret diffusion; *n*-Butane/air flame; Fickian diffusion; Flame speed

1. Introduction

Compared with Fickian diffusion, Soret diffusion caused by the temperature gradient is often seen as the secondary mass diffusion, and it is expected to be very important in the presence of large temperature gradient close to the intense reaction zone. As we known Soret effect drives light species toward the hot active reaction zone and heavy species away from this hot zone [1,2]. Furthermore, at detailed flame structure level, the Soret effect of fuels would control the total enthalpy of reaction zone, and Soret effect of intermediates would strongly affect the reaction dynamics.

Ern and Giovangigli [3,4] studied the effects of Soret effect in freely-propagating planar flame speed in hydrogen/air and methane/air flames. It was indicated that Soret diffusion very slightly lowers the laminar flame speeds on the lean side, but slightly elevates them on the rich side in hydrogen/air flame and there is almost no effect

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of Soret diffusion on the speed of methane/air flame. They also studied the counterflow premixed flames, and it was shown that Soret diffusion extends the stretch-affected extinction limit of lean flames while the corresponding limit of rich flames is narrowed in hydrogen/air flame. Detailed computational studies on the structure and extinction of nonpremixed counterflow methane/air flames were conducted by Williams [5], it was found that the extinction stretch rate is extended when Soret diffusion was included. In our former study [6,7], we systematically studied the separate and combined effects of Soret diffusion of the hydrogen molecule (H_2) and radical (H) on the structure and flame speed of the freely propagating planar premixed flames as well as the strain-induced extinction response of premixed and nonpremixed counterflow flames in hydrogen/air flame. It was shown that for freely propagating planar premixed flames, the Soret diffusion of H_2 is insensitivity to the flame speed and the Soret diffusion of H leads to obvious reduction of flame speed. For symmetric counterflow premixed flames and one side with cold nitrogen counterflow premixed flames, Soret effect of H_2 has much greater influence on reducing the extinction strain rate in rich flames and extending extinction strain rate in lean flames. For counterflow diffusion flames, the Soret effect of H_2 and H both substantially extend the extinction strain rate, and Soret effect of H has stronger effect.

However, to our knowledge, for heavy fuel, the relative influences of Soret effect on flame structures, flame speed and strain-induced extinction have not been reported previously at detailed flame structures level in the literature. At flame sheet level, Arias–Zugasti and Rosner [8] studied the Soret effect of heavy fuels in counterflow diffusion flames, and found that flame temperature is decreased and the position of flame shift to the fuel side with Soret effect of heavy fuels. For the very heavy species in the 'particle' limit, there are some studies on the influence of Soret effect on soot formation [9] and metal oxides [10,11] in flames. In this paper, for deeply revealing the detailed physical mechanism of Soret effect on the flame structure and propagation as well as strain-induced extinction in heavy fuel flames, based on the computational method the freely-propagating premixed flames is studied with and without Soret effect.

2. Problem specification

The one dimensional freely-propagating planar premixed flame was simulated using the flame code developed by Kee et al. [12], and the continuity equation and the conservation equations for energy and species as well as equation of state are:

Continuity:

$$m = \rho u \tag{1}$$

Energy:

$$m\frac{dT}{dx} - \frac{1}{c_p}\frac{d}{dx}\left(\lambda\frac{dT}{dx}\right) + \frac{1}{c_p}\sum_{k=1}^{K}\rho Y_k V_k c_{pk}\frac{dT}{dx} + \frac{1}{c_p}\sum_{k=1}^{K}\omega_k h_k W_k = 0 \tag{2}$$

Species:

$$m\frac{dY_k}{dx} + \frac{d}{dx}(\rho Y_k V_k) - \omega_k W_k = 0 \tag{3}$$

Equation of state:

$$\rho = \frac{p\overline{W}}{RT} \tag{4}$$

where x denotes the spatial coordinate, m the mass burning rate per unit area, ρ the mass density, u the velocity of the mixture, T the temperature, c_p the constant-pressure heat capacity of the mixture, λ the thermal conductivity of the mixture, K the total number of species, Y_k the mass fraction of the kth species, V_k the diffusion velocity of the kth species, ω_k the molar rate of production per unit volume, h_k the specific enthalpy, W_k the molecular weight of the kth species, p the pressure, R the universal gas constant. The Hydrogen oxidation mechanism involving 9 species and 21 reactions was taken from Li et al.[13] and the *n*-butane oxidation mechanism is that of Wang et al. [14] involving 111 species and 784 reactions. Transport properties of *n*-butane/air flames were evaluated by an updated transport database, Sandia transport Interpreter and subroutine library of Wang et al.[14] and multicomponent formulation was used in this study.

3. Results and discussion

3.1. Laminar premixed flames speed with Soret effect

In Fig. 1, the computed laminar flame speeds are compared with the measured results [15] for *n*-butane/air flames with and without Soret effect at atmospheric condition. It is shown that the computational results with and without Soret effect basically agree with measurement values and Soret effect has only minute effects on the laminar flame speed for *n*-butane/air flames. It is known that the *n*-butane/air flames include not only very light species such as H radical and H_2, but also heavy species such as fuel *n*-butane. Due to the Soret effect, the temperature gradient drives the very light and heavy species and modifies the concentration and distribution of these species, which in turn affects

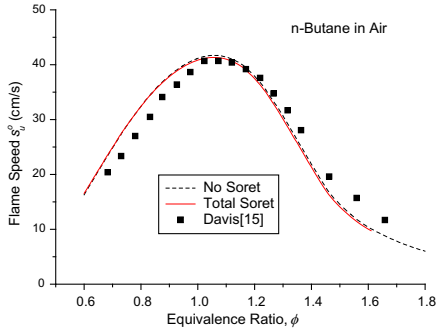

Fig. 1. (lines) Predicted and (symbols) measured speeds of laminar premixed *n*-butane/air flames. Predictions include no soret and total soret effects for all species.

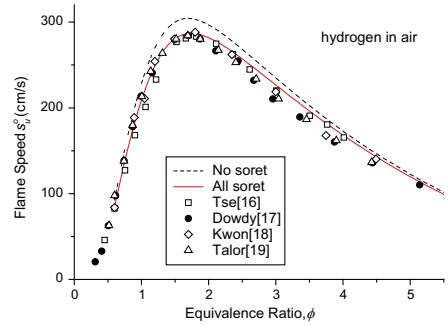

Fig. 3. (lines) Predicted and (symbols) measured speeds of laminar premixed hydrogen/air flames. Predictions include no soret and total soret effects for all species.

the individual reaction rates and flame speed. However, as result shown, for the *n*-butane/air flame speed, the Soret effect almost has no influence. For finding the key active radicals in the flame propagation, the sensitivity analysis of flame speed is studied in Fig. 2 which shows the ranked logarithmic response sensitivity coefficients of flame speed for *n*-butane/air at the equivalence ratios of 0.8, 1.0, and 1.4. As is known, the two most important reactions in a hydrocarbon combustion process are $H + O_2 = O + OH$ and $CO + OH = CO_2 + H$, and flame speed of *n*-butane/air are indeed the most sensitive to these two reactions. It is seen that these ranked key reactions do not involve any of the heavy species such as fuel *n*-butane and most of them involve the very light species H radical. Thus only the Soret effect of light species H radical should plays the most import role in flame speed of *n*-butane/air.

It is noted that Soret effect of H almost has no influence for the flame speed of *n*-butane/air, but plays a crucial role in reduction of flame speed for hydrogen/air flames [6,7]. For comparison between *n*-butane/air and hydrogen/air flames, we plot Fig. 3, in which the computed laminar burning velocities are compared with the measured results [16–19] for hydrogen/air flames with and without Soret effect at atmospheric condition. It is seen that the computational values without Soret effect from fuel lean to rich are uniformly larger than the measured values and the computed flame speed with Soret effect for all species agrees closely with the measurement values. This means Soret effect has prominent effects on the laminar flame speed for hydrogen/air flames. Actually, in our former study [6,7], based on the detailed chemical structures analysis we have done the deeply study for Soret effect in hydrogen/air flames and found that due to the "downstream transport" of H radical, the Soret diffusion of H leads to the observed decrease in the hydrogen/air laminar flame speed in Fig. 3.

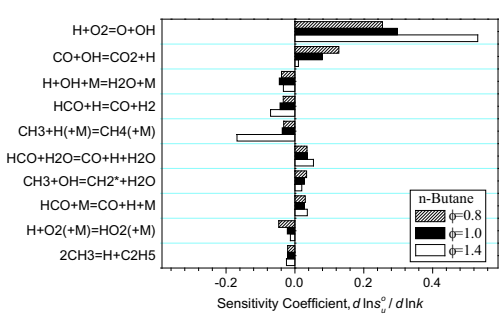

Fig. 2. Logarithmic response sensitivity coefficients of flame speed computed for *n*-butane/air at the equivalence ratios of 0.8, 1.0, and 1.4.

3.2. Soret diffusion flux of H radical

A direct and satisfactory explanation for the difference of H Soret effect between *n*-butane/air and hydrogen/air flames requires an examination of fickian and Soret diffusion flux of H, and need a comparison of H diffusion flux between *n*-butane/air and hydrogen/air flames. Figure 4 shows the spatially-resolved Fickian and Soret diffusion fluxes of H for *n*-butane/air flame at $\phi = 1$. It is seen that compared with Fickian diffusion flux of H, the value of H Soret diffusion flux is too small to exert an influence on the total diffusion flux of H. For comparison, we also plot Fig. 5, Spatially-resolved Fickian and Soret diffusion fluxes of H for hydrogen/air flame. Comparing Fig. 4 with Fig. 5 for the order of magnitude in diffusion flux of H, the results show that compared with hydrogen/air flame, the large reduction of H Soret diffusion flux in *n*-butane/air flame is the crucial reason for that Soret effect of H almost has no influence in the flame speed. We plot Fig. 6 to clearly show this result. For revealing the detailed physical mechanism of this large reduction in H Soret diffusion flux of *n*-butane/air flame, in next section, the "Driving

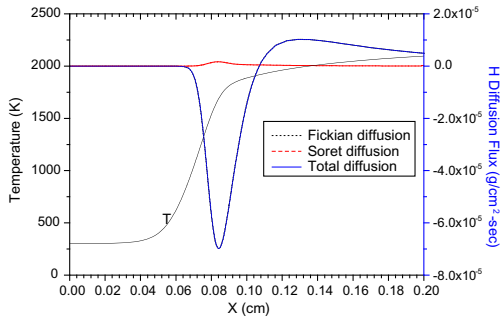

Fig. 4. Spatially-resolved Fickian and Soret diffusion fluxes of H for the laminar premixed *n*-butane/air flame at $\phi = 1$ and atmospheric conditions, allowing Soret diffusion for all species; temperature distribution superimposed for reference.

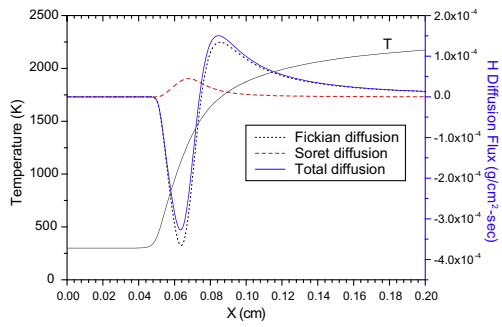

Fig. 5. Spatially-resolved Fickian and Soret diffusion fluxes of H for the laminar premixed hydrogen/air flame at $\phi = 1$ and atmospheric conditions, allowing Soret diffusion for all species; temperature distribution superimposed for reference.

force" and coefficients of H Soret diffusion as well as their spatial relation are detailedly studied.

3.3. "*Driving force*" *and mole fraction of H*

The species diffusion flux consists of two terms: fickian and Soret diffusion flux. Therefore:

$$j_k = -\rho D_k \nabla Y_k - D_k^T \frac{1}{T} \nabla T. \tag{5}$$

where, j_k is diffusion flux of kth species, ρ is the density of the mixture, D_k is fickian diffusion coefficient of kth species, Y_k is the mass fraction of kth species, D_k^T is the Soret diffusion coefficient of kth species and T is the temperature of the mixture. The first term on RHS of Eq. (5) is fickian diffusion flux caused by the concentration gradient and the second term is Soret diffusion flux caused by the "driving force" gradT/T. We use multicomponent formulation to evaluate the transport properties, so the Soret diffusion coefficients are given by [11]

$$D_k^T = -\frac{8 m_k X_k}{5R} a_{k00}^1. \tag{6}$$

Where m_k is molecular mass of kth species, R is universal gas constant and X_k is mole fraction of kth species. a_{k00}^1 is computed from the solution of a system of equations defined by L matrix and this system is given by

$$\begin{pmatrix} L^{00,00} & L^{00,10} & 0 \\ L^{10,00} & L^{10,10} & L^{10,01} \\ 0 & L^{01,10} & L^{01,01} \end{pmatrix} \begin{pmatrix} a_{00}^1 \\ a_{10}^1 \\ a_{01}^1 \end{pmatrix} = \begin{pmatrix} 0 \\ X \\ X \end{pmatrix} \tag{7}$$

The components of the L matrix are given by Dixon–Lewis [20]. It is seen that Soret diffusion coefficient D_{k00}^1 should vary with mole fraction X_k in the same manner. Figures 7 and 8 show the spatially-resolved mole fraction and Soret diffusion coefficients of H respectively and the results shows they indeed vary in the same manner.

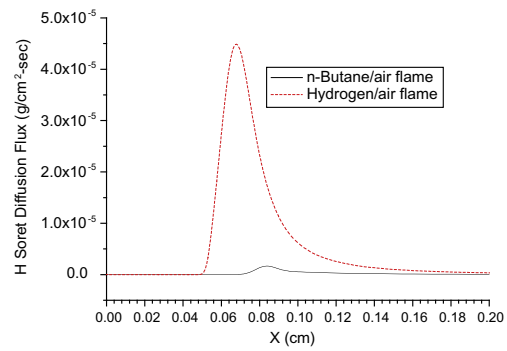

Fig. 6. Spatially-resolved Soret diffusion fluxes of H for the laminar premixed *n*-butane/air flame and hydrogen/air flame at $\phi = 1$ and atmospheric conditions, allowing Soret diffusion for all species.

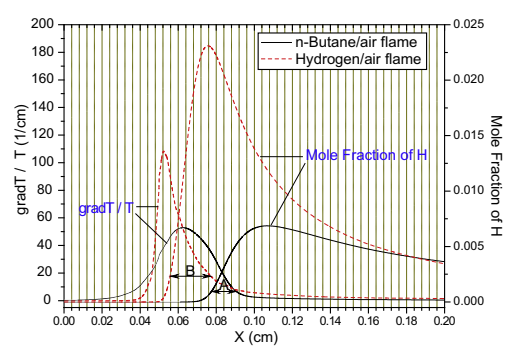

Fig. 7. Spatially-resolved mole fraction of H for the laminar premixed *n*-butane/air flame and hydrogen/air flame at $\phi = 1$ and atmospheric conditions; temperature gradient superimposed for reference. Horizontal arrows indicate the overlap region of gradT/T and mole fraction of H for n-Butane/air flame (A) and Hydrogen/air flame (B).

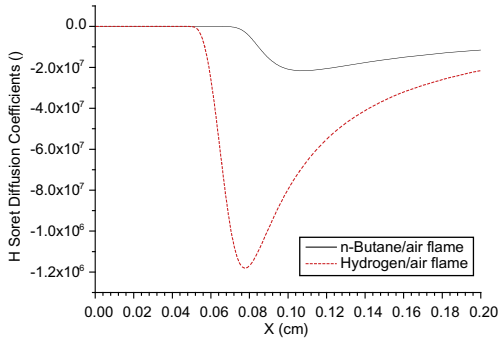

Fig. 8. Spatially-resolved H Soret diffusion coefficients for the laminar premixed *n*-butane/air flame and hydrogen/air flame at $\phi = 1$ and atmospheric conditions.

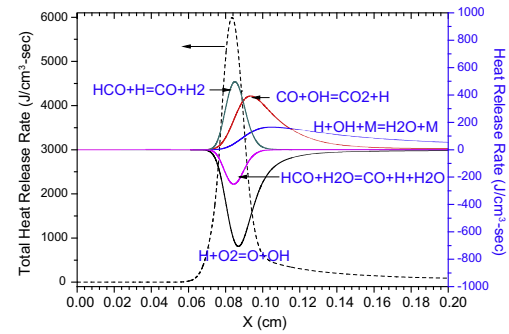

Fig. 9. Spatially-resolved total heat release rate and heat release rate of key exothermic and endothermic reactions for *n*-butane/air flame at $\phi = 1$ and atmospheric conditions.

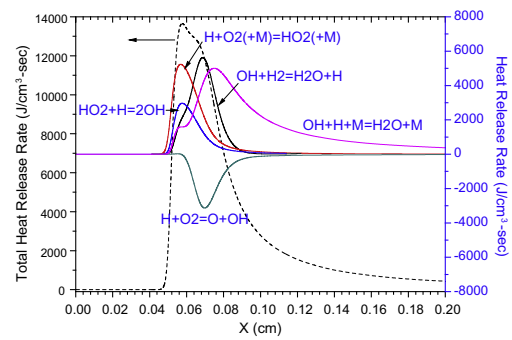

Fig. 10. Spatially-resolved total heat release rate and heat release rate of key exothermic and endothermic reactions for hydrogen/air flame at $\phi = 1$ and atmospheric conditions.

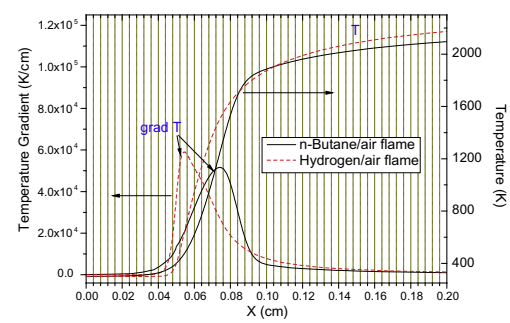

Fig. 11. Spatially-resolved temperature and temperature gradient for the laminar premixed *n*-butane/air flame and hydrogen/air flame at $\phi = 1$ and atmospheric conditions.

Figure 7 shows that the values of gradT/T and mole fraction of H for *n*-butane/air are both lower than hygrogen/air flames, but this obviously cannot lead to the large reduction of H Soret diffusion flux for *n*-butane/air flame in Fig. 6. The Soret diffusion flux not only depends on the values of "driving force" and mole fraction, but it also depends on the relation of their spatial locations. In Fig. 7, It is seen that the overlap region of gradT/T and mole fraction of H for *n*-butane/air flame spans between $x \approx 0.072$–0.10 cm, which is a very small region and cannot includes the large values of gradT/T and mole fraction of H. However, for the hydrogen/air flame the overlap region spans between $x \approx 0.051$–0.14 cm, and the peaks of both gradT/T and mole fraction of H are embedded within this broad region. It is also noted that this overlap region is precisely the region of Soret diffusion flux of H in Fig. 6. Thus, the too narrow overlap region of "driving force" and mole fraction of H is the reason for that Soret effect of H almost has no influence in the *n*-butane/air flame speed.

A satisfied explanation of this narrow overlap region for *n*-butane/air flame requires the examination of chemical structure and comparison with hydrogen/air flame. Figures 9 and 10 show the spatially-resolved heat release rate for *n*-butane/air and hydrogen/air flame. It is seen that for *n*-butane/air flame the total heat release rate start to increase at $x \approx 0.06$ cm, reaching its maximum values around $x \approx 0.084$ cm and for hydrogen/air flame this region which spans between $x \approx 0.044$–0.056 cm is smaller than *n*-butane/air flame. Thus, in Fig. 11 the peak of temperature gradient is located at the region with higher temperature of 1335.4 K for *n*-butane/air flame and the lower temperature is 632.4 K for hydrogen/air flame. Compared with hydrogen/air flame, the higher temperature within the whole region of gradT largely reduces the "driving force" gradT/T for *n*-butane/air flame, especially on the rear part of gradT/T profile. Compared with the location of peak of temperature gradient, the peak of gradT/T shifts to unburned side in *n*-butane air flame due to such reduction. However due to the location of temperature gradient with lower temperature, the reduction of gradT/T on the rear part of its profile and the location shift of peak

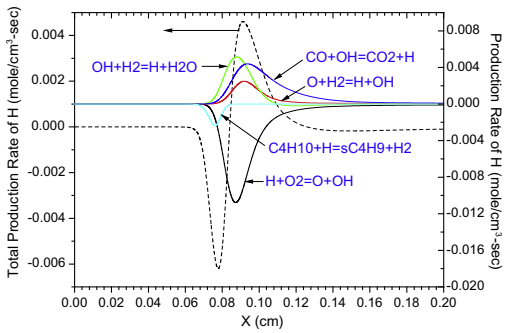

Fig. 12. Spatially-resolved total production rate of H and production of H for key H production and consumption reactions for *n*-butane/air flame at $\phi = 1$ and atmospheric conditions.

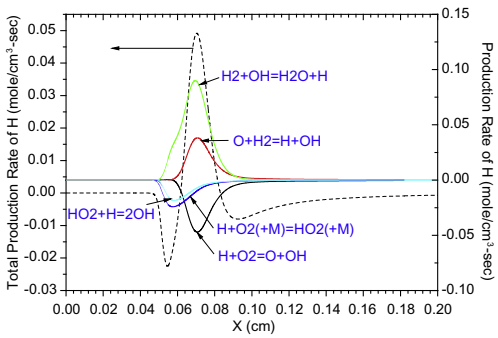

Fig. 13. Spatially-resolved total production rate of H and production of H for key H production and consumption reactions for hydrogen/air flame at $\phi = 1$ and atmospheric conditions.

are not obvious for hydrogen/air flame. Figures 12 and 13 respectively show computed profile of production rate of H for *n*-butane/air and hydrogen/air flame. It is seen that compared with hydrogen/air flame the very large reaction rate of crucial chain-branching reaction $H + O_2 = OH + O$ leads to the broader consumption layer of H which spans between $x \approx 0.06$–0.084 cm in *n*-butane/air flame, furthermore this broad consumption layer of H weakens the mole fraction of H in the front of its spatial profile. Compared with hydrogen/air flame, the reason for the narrow overlap region in *n*-butane/air flame has two factors. First, the high temperature region reduces the "driving force" on the rear part of its profile. Second, the broad consumption layer of H weakens the mole fraction of H in the front part of its spatial profile.

4. Conclusions

In the present investigation, compared with hydrogen/air flame we have study the influence of Soret effect on the freely-propagating planar premixed flame speed of *n*-butane/air mixtures. The Soret diffusion flux of the kth species just exists in the overlap region which consists of the rear part region of "driving force" gradT/T and the front part region of mole fraction of the kth species in the freely-propagating planar premixed flame. For *n*-butane/air flame, the total heat release rate rise slowly to the peak, so the most part of temperature gradient locates in the region with high temperature which largely reduces the "driving force" especially on the rear part of its profile and compared with the peak of temperature gradient the location of "driving force" peak is closer to the unburned side. On the other hand, The key H consumption reaction $H + O_2 = OH + O$ leads to the broader consumption layer of H which weakens the front part of mole fraction profile of H and makes the peak of H mole fraction far away from the unburned side. Thus, the overlap region is narrow and low values of "driving force" and mole fraction of H embed in it. Compared with Fickian diffusion flux of H, the Soret diffusion flux of H is too small to influence the total diffusion flux in *n*-butane/air flame. So, the Soret effect has almost no influence on the flame speed of *n*-butane/air mixtures.

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